



Milky Way's sibling found

Scientists have found the remnants of a long lost sibling of the Milky Way galaxy.
<https://digit.in/sep18-74-1>



New frog identified

Sylvia's Tree Frog has been identified as a new species after almost 20 years of research
<https://digit.in/sep18-74-2>

A Golden Orb spider weaves a web on board the ISS

Strange ISS Experiments

Credit: NASA

The microgravity environment on the International Space Station (ISS) allows a number of experiments to be conducted, that are not possible on the Earth.

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FLATWORMS AND SPIDERS

Flatworms have the capability of regenerating parts of their bodies that have been chopped off. To understand how gravity and the geomagnetic fields affects the process of regeneration, scientists chopped off the heads and tails of flatworms, sealed them in water pouches, and sent them up to the ISS. On their return the re-growth was analysed and compared to that of flatworms that had never left the Earth. The flatworms showed a number of changes on their return, including changes in their microbiomes, and showed an increase in the variety of preferences to light levels. One of the flatworms sent to space had been chopped in half, which typically results in two

whole flatworms. In space, however, the flatworm grew two heads instead of one – one at each end of its body. Such a thing had never been observed on Earth before. The flatworm continued to make two heads as it further divided back on Earth.

Spiders in Space was an educational experiment where Golden Orb spiders were made to spin their webs in space. A live stream from the ISS allowed students to compare the spiderwebs to the same species of spiders on the ground. The experiment found that the shape of the spider webs did not change much in space. They were more circular in comparison to the webs of the same species on Earth though.

ASTROBIOLOGY

If you map the increasing complexity of life, then it is a straight line that

goes back to before the formation of the Earth. This is just one of the many pieces of evidence that suggests that life across the universe could be a result of life travelling in a dormant form between celestial bodies – piggybacking on rocky debris. The panspermia hypothesis, as outlandish as it may seem, has actually been tested on board the ISS. The experiment is known as The Astrobiology Exposure and Micrometeoroid Capture Experiments (Tanpopo), and is the first astrobiology experiment to be conducted on board the ISS. Blocks of a low density material called aerogel are exposed to space, to collect tiny particles. The particles include high altitude terrestrial particles as well as micrometeoroids, which could harbour organics. At the same time, microbes from Earth, extremophiles and organic compounds are exposed to the space environment. These aerogel blocks are later retrieved, and the particles evaluated. The analysis of the particles are still ongoing, but there is some early evidence to suggest that at least some microorganisms can survive